

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

20

Seat Locking Bracket

256 × 226 × 31 mm · 71.1 cm³ · AI 0.192 kg / Steel 0.558 kg

Manufacturability: 20/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	255.8 × 226.1 × 31.0 mm	Surface Area	919.4 cm ²
Volume	71.07 cm ³	Weight (AI)	0.192 kg
Weight (Steel)	0.558 kg	Weight (Plastic)	0.075 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Bend	15	High	15 bend operations inferred from numOps=12 station progressive die (1 blank + 15 bends)
Threaded feature	1	Medium	Threaded holes detected — requires secondary tap/insert operation after forming
Cylindrical hole/lance	50	Medium	Small radius cylindrical features (r 1.9–7.6mm) consistent with pierced/lanced holes in stamped bracket
Free-form flange/lip	235	Medium	High B-spline face count (235/426) suggests curved flanges, lightening cutouts and locking-tab profiles rather than a cast/moulded surface
Undercut/re-entrant features	32	High	32 undercut faces relative to Z draw direction — consistent with formed locking hooks/tabs typical of a seat-lock bracket, manageable in stamping via die action, not relevant to casting draft

§3 — Material Analysis

AI-suggested

Primary Material:

DC01 Cold-Rolled Mild Steel Sheet (62% confidence)

Part name 'Seat Locking Bracket' indicates a safety-critical automotive seating structural/locking component. Such brackets are almost universally stamped from cold-rolled or high-strength steel sheet (not aluminium) for strength, fatigue life and low cost at high volume (200k/yr). No aluminium signal (no thin uniform HPDC wall, no lightweighting note) is present, so per guidance we default to steel. DC01 mild steel sheet is the standard low-cost stamping grade; a higher-strength steel (e.g. HSLA/DC04) may be substituted if higher crash/lock loads are required.

Alternatives:

1045 Medium Carbon Steel (higher strength stamping/forming grade) (25%) · Aluminium 6061 Sheet (lightweighting alternative) (10%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Progressive die stamping	sheet_metal	90%	0.0005	User-forced primary process; 200,000 units/yr strongly favours a hardened progressive die running blank + 15 bend stations, matching the 12-station tooling and 34 strokes/min feed-limited rate provided.
Sheet metal fab (laser cut + brake press)	sheet_metal_fab	15%	0.0500	Only viable at low volumes/prototyping; uneconomical at 200k/yr due to per-part labour cost versus a 25k GBP progressive die.

§5 — Manufacturability Risks (Score: 20/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (32 faces)	32 undercut/re-entrant faces relative to draw direction indicate complex locking-hook geometry that will require multiple forming stations, cam/side-action tooling in the progressive die, or secondary forming presses.	Consolidate hook/lock features into fewer forming stages; review geometry for die-friendly bend sequencing to avoid tearing/wrinkling.
High	Minimum wall/gauge (0.53mm local)	Localised thin sections (min 0.53mm vs mean 1.55mm) suggest tight radii or lanced tabs that are prone to cracking or excessive thinning during forming at production gauge.	Increase local bend radii to $\geq 1 \times t$ and minimum flange lengths to avoid split/crack defects; consider a coining or ironing station if thinning is functional.
Medium	15 bend operations	High bend count increases die complexity, springback variation and cumulative dimensional stack-up across the strip.	Group compatible bends per station, add spring-back compensation angles, and validate with FEA before die cut.
Medium	Threaded feature	Threads cannot be formed in-die at this gauge; requires a secondary tapping or self-clinching fastener insertion operation, adding cost/cycle time.	Use flow-form tapping, PEM inserts, or weld nuts added as a secondary automated station.

§6 — DFM Issues (sheet_metal)

Severity	Area	Description	Impact	Fix
High	Locking hook / undercut region	32 undercut faces relative to the vertical draw direction imply hooked/re-entrant lock geometry that a straight-punch progressive die cannot form in a single stroke.	Requires cam-action or multi-stage forming stations, adding 15-20% to die build cost and cycle complexity; risk of part hang-up/jamming in strip.	Redesign lock feature for single-direction forming or add a dedicated cam/wedge station in the die design.
High	Thin local gauge (0.53mm min)	Local wall/flange thickness far below the mean 1.55mm suggests tight radius bends or lanced tabs at risk of splitting during forming at 2.0mm nominal stock.	Scrap/rework rate increase, particularly at high-volume (200k/yr) production; potential warranty risk on a safety-critical lock part.	Increase minimum bend radius to $\geq 1 \times \text{material thickness}$ and validate flange length against $\geq 3 \times$ minimum rule; run strain analysis (FLD) on critical bends.
Medium	15-bend die (numOps=12)	High station count increases progressive die length, cumulative springback error and strip-feed accuracy requirements at 34 strokes/min.	Longer die build/tryout time and higher risk of dimensional drift over the die's 1,000,000-hit life.	Combine non-interfering bends into common stations; add pilot/restrike stations for critical angles; use bend-relief slots to control springback.
Medium	Threaded feature	Threads cannot be produced directly in the stamping die and require an off-line tap or fastener-insertion operation.	Adds secondary machine/labour cost (~0.05hr/part) and a second QC checkpoint.	Consider self-clinching PEM nuts inserted in-die (press-fit station) to eliminate the separate tapping operation.

Severity	Area	Description	Impact	Fix
Low	Strip layout utilisation	Blank 269×237mm on a 272mm pitch / 243mm strip width gives moderate material utilisation typical of an irregular bracket outline.	Scrap skeleton adds ~10-15% to raw material cost per part.	Nest mirrored parts two-up across strip width if bracket geometry allows, to raise utilisation and reduce coil cost per piece.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£1.05	£1.60	£2.55

Net Weight	0.558 kg	Material	mat-dc01
Recommended Process	sheet_metal	Cycle Time	0.0505 hr/part
Setup Time	0.750 hr	Operations	2 ops
Operations Detail	Progressive die blank & 15-station forming (mach-haas-vf2, 0.0005 hr) Secondary tap/insert for threaded feature (mach-drill, 0.0500 hr)		

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£0
Die Life	0 shots
Cavities	0
Yield	0.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation

The part is analysed as a stamped sheet-metal seat locking bracket per the user-forced process. Geometry (0.0396 fill ratio, extensive B-spline/free-form face content and 32 'undercuts') reflects a fabricated formed-sheet part rather than a cast or moulded one — the undercuts are simply re-entrant locking-hook bends relative to the chosen Z draw direction, not casting draft violations. Deterministic blank (269×237mm), pitch (272mm), strip width (243mm), 12-station numOps and 34 strokes/min feed rate are used verbatim from the geometry-derived sheet-metal analysis. Material is assigned to steel (DC01 cold-rolled sheet) rather than aluminium because the part name and function (a seat lock, a safety-critical automotive structural component) strongly favour steel per the material-selection guidance, absent any aluminium-specific signal. Net weight (0.558kg) uses the steel density figure supplied. Thickness (2.0mm), die cost/life/type and shear strength are estimated engineering judgements flagged as UNDECIDED inputs, chosen to be consistent with typical seat-bracket stamping gauges and progressive-die tooling for 200k/yr volumes; the 25,000 GBP progressive die cost is used verbatim from the parametric estimate. Manufacturability score of 20/100 is carried through unchanged, reflecting the high complexity (undercuts, thin sections, many bends) even though the part is being made by an appropriate (stamping) process.

§9 — Analysis Limitations & Assumptions

1. Material (steel vs aluminium) and gauge thickness are not stated in the geometry file and are inferred from part name/function per user guidance — actual drawing/spec should confirm.
2. Progressive die cost/life figures are parametric estimates, not quoted from an actual toolmaker for this specific 15-bend geometry.
3. OCCT volume/surface figures were derived from a shell/wall-thickness reconstruction rather than a true flat-pattern sheet-metal model, so blank nesting/utilisation figures are approximate.
4. Undercut and thin-gauge feature counts are inferred from geometric analysis relative to an assumed Z draw direction and may not reflect the true forming direction defined by the actual die design.

Stage-1 pre-selection: sheet_metal (100%) ·